

The Real AI Divide Isn't Intelligence. It's Bandwidth.

How AI changes capacity more than capability

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ABSTRACT

AI is often described as a productivity tool. I think it changes something more fundamental: the amount of cognitive bandwidth people have available. This memo explores why the real divide in AI adoption may have less to do with intelligence than with the conditions under which people are asked to learn.

There's a growing assumption that AI will divide people based on who understands it first. Who knows how to use it. Who can write better prompts. Who can keep up.

That doesn't quite match what I'm seeing.

The divide feels less like intelligence than bandwidth. Not everyone moves through the day with the same amount of cognitive margin. Some people have enough room to experiment. They can spend an hour exploring a new tool without worrying whether it was the most efficient use of their time.

They can tolerate the awkwardness that comes with learning something unfamiliar.

AI is often described as saving time. I think a better way to describe it is that it reduces the effort required to translate thought into something usable. It can help organize ideas, draft communication, summarize information, and create structure where there wasn't any before. In other words, it reduces part of the translation work that many knowledge systems quietly require.

That's a meaningful shift. It doesn't make people more capable. It changes how much of their existing capability they can bring into the room. But benefiting from that shift still requires enough bandwidth to explore the tool in the first place. And that's the part I think we underestimate.

Bandwidth isn't distributed randomly. It's shaped by circumstances. Economic pressure. Health. Workload. Caregiving. Administrative burden. Institutional complexity. Under those conditions, a new tool doesn't necessarily feel like an opportunity. Sometimes it just feels like another thing asking for attention.

Stress itself narrows attention. It makes experimentation more expensive. It encourages short-term decisions over long-term investment. The people who could benefit most from additional cognitive support may also be the people with the least capacity to adopt it.

That's why I don't think the most important question is whether AI makes people more productive. The more interesting question to me is whether it expands **participation**.

Does it make it easier for different kinds of minds to contribute?

Does it reduce the amount of translation required to participate in modern knowledge work?

Or does it mostly accelerate the people who already had enough room to adapt, increasing the distance between those who have cognitive margin and those who don't?

That feels like a research question worth pursuing.